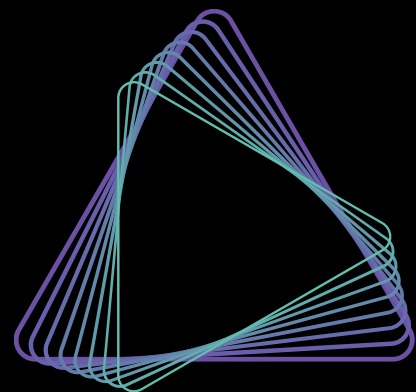




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LIFE SCIENCES

Bulk CRISPR Screen 分析培训

2024年08月

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所需安装的软件

01

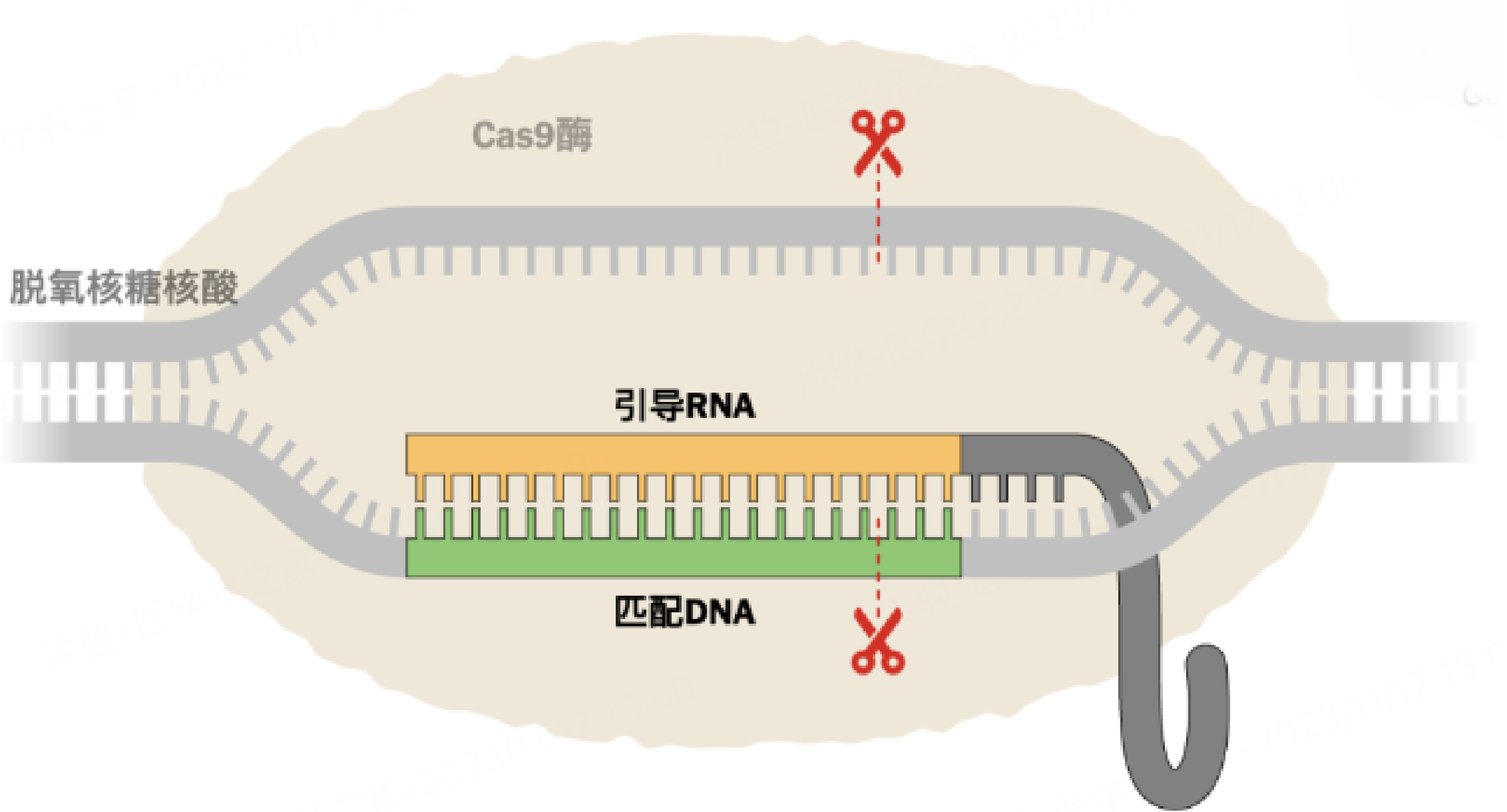
1.1 软件列表

- mageck
- fastqc
- cutadapt
- R(包括MAGeCKFlute、tidyr、ggplot2、stats等包)

什么是CRISPR Screen

02

2.1 基因编辑技术



1. 瞄准正确的基因

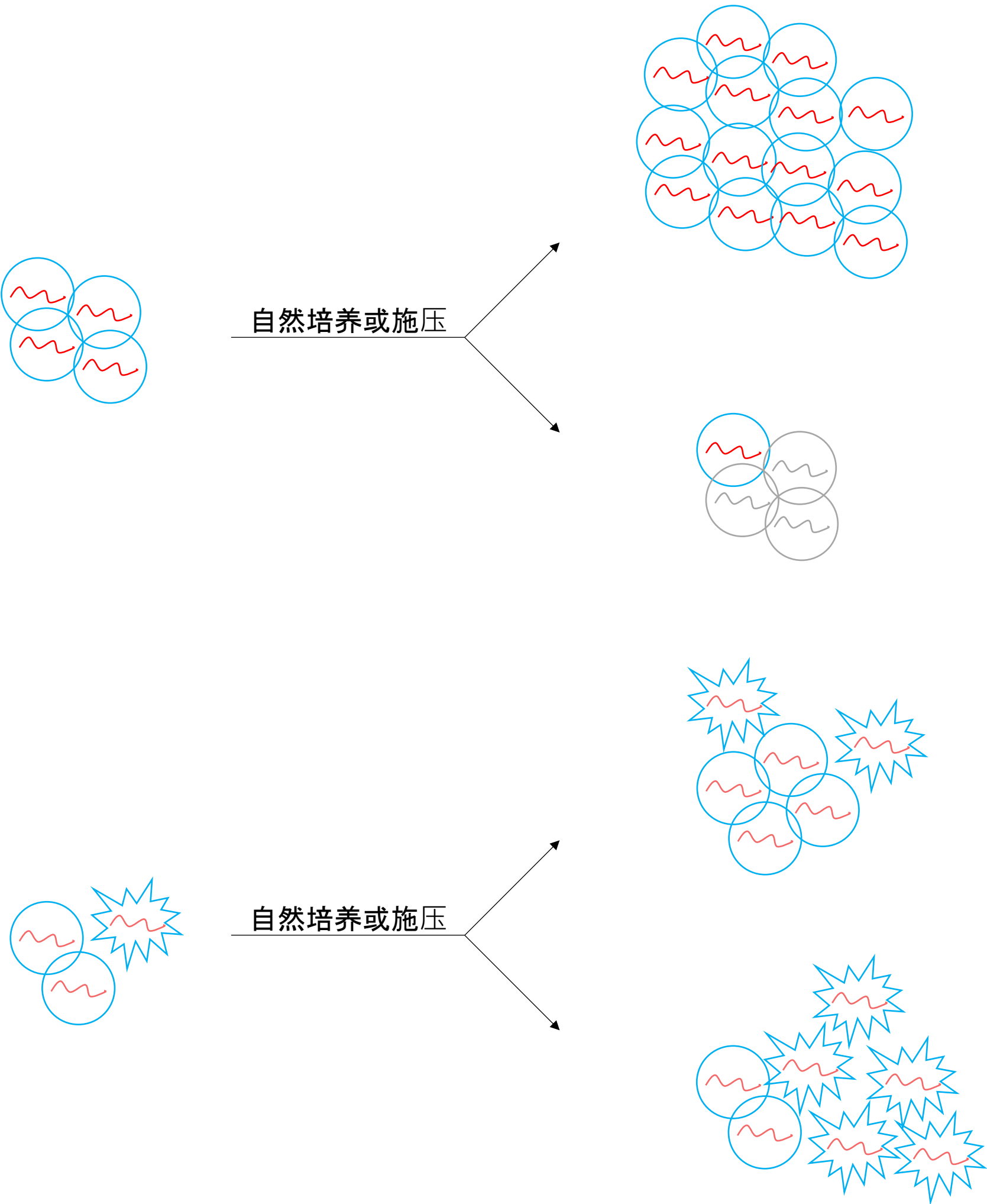
科学家设计了一段与他们想要编辑的DNA相匹配的RNA。这称为**引导RNA**。

2. 绑定目标

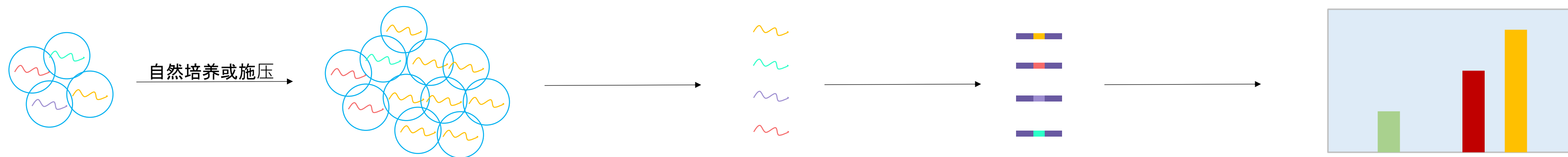
一种名为**Cas9**的酶与一段DNA结合，并暂时解开一段DNA。

3. 切割DNA

如果引导RNA与DNA的一部分相匹配，Cas9酶就会切割DNA双螺旋的两条链。



2.2 CRISPR Screen



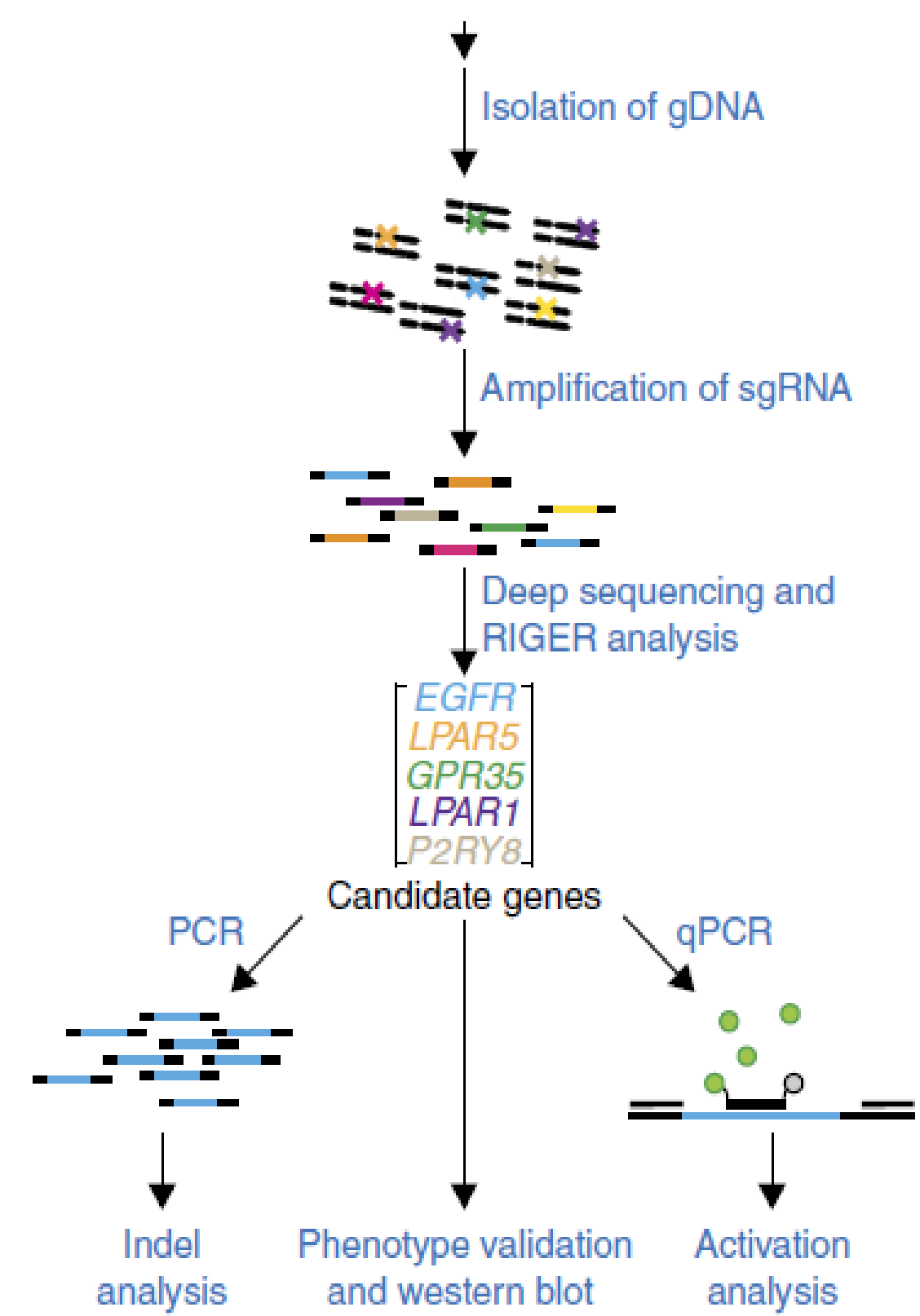
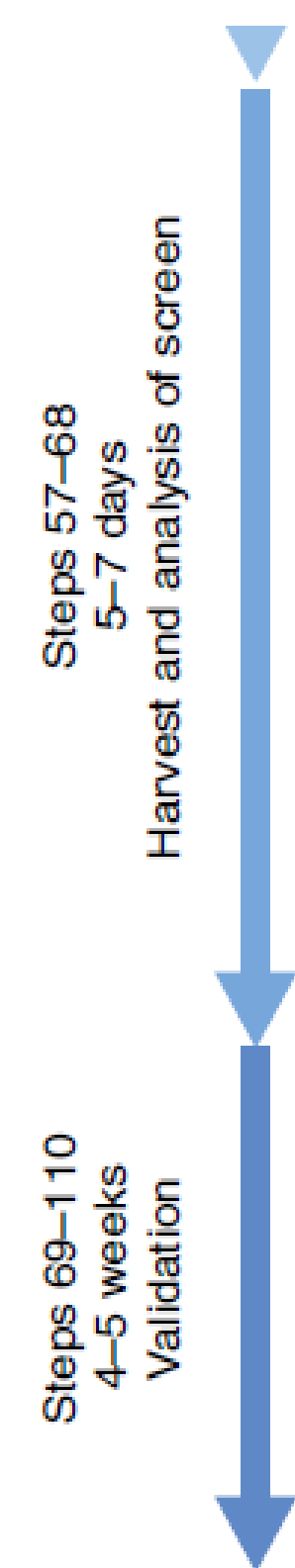
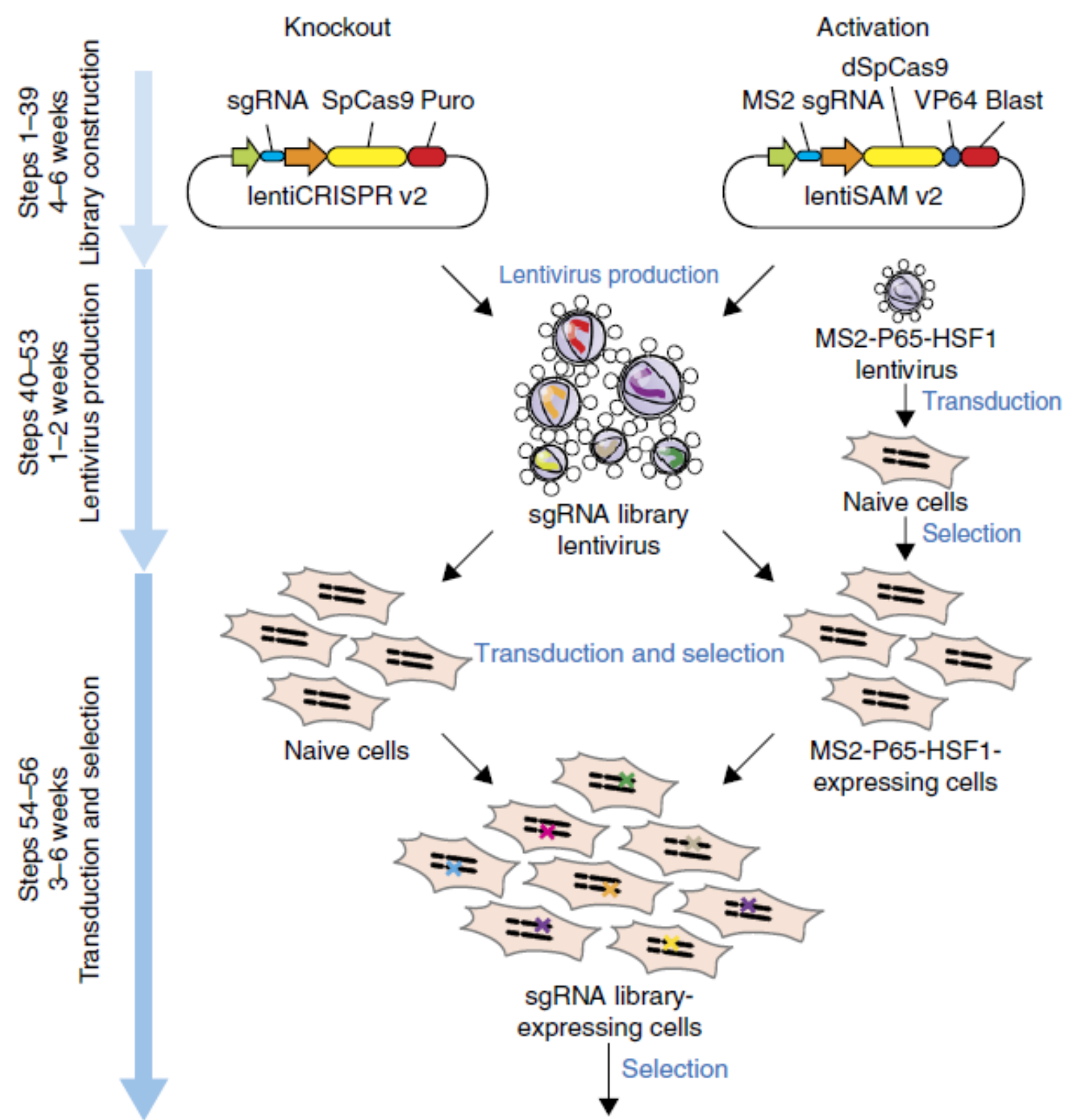
CRISPR Screen，也被称为全基因组CRISPR或高通量基因编辑技术。

基因编辑结合NGS测序和生物信息学分析，量化靶向目标基因的sgRNA在筛选前后的变化，预测符合条件的候选基因。

怎么做CRISPR Screen

03

3.1 总体流程



设计文库 — 包装病毒 — 转染细胞 — 施压筛选 — 提取核酸 — 建库测序 — 数据分析 — (实验验证)

3.2 数据准备

数据准备

```
mkdir -p ~/BulkCrispr
cd ~/BulkCrispr
WorkDir=$(pwd)
mkdir -p ${WorkDir}/00.RawData
cd ${WorkDir}/00.RawData
ln -sf /CRISPR/BulkCrispr/00.RawData/Baseline.R1.fastq.gz
ln -sf /CRISPR/BulkCrispr/00.RawData/Baseline.R2.fastq.gz
ln -sf /CRISPR/BulkCrispr/00.RawData/Control.R1.fastq.gz
ln -sf /CRISPR/BulkCrispr/00.RawData/Control.R2.fastq.gz
ln -sf /CRISPR/BulkCrispr/00.RawData/Treatment.R1.fastq.gz
ln -sf /CRISPR/BulkCrispr/00.RawData/Treatment.R2.fastq.gz
cd ${WorkDir}
```

样本介绍:

Baseline: 转染初始时提取的样本 (Day0)

Control: 转染后培养了7天的样本 (Day7)

Treatment: 转染后培养了14天的样本 (Day14)

```
(base) crispr001@iZbp18758pdxsp0rsgmsuhZ:~/BulkCrispr$ ls -l 00.RawData/
total 0
lrwxrwxrwx 1 crispr001 crispr001 50 Aug 14 11:14 Baseline.R1.fastq.gz -> /CRISPR/BulkCrispr/00.RawData/Baseline.R1.fastq.gz
lrwxrwxrwx 1 crispr001 crispr001 50 Aug 14 11:14 Baseline.R2.fastq.gz -> /CRISPR/BulkCrispr/00.RawData/Baseline.R2.fastq.gz
lrwxrwxrwx 1 crispr001 crispr001 49 Aug 14 11:14 Control.R1.fastq.gz -> /CRISPR/BulkCrispr/00.RawData/Control.R1.fastq.gz
lrwxrwxrwx 1 crispr001 crispr001 49 Aug 14 11:14 Control.R2.fastq.gz -> /CRISPR/BulkCrispr/00.RawData/Control.R2.fastq.gz
lrwxrwxrwx 1 crispr001 crispr001 51 Aug 14 11:14 Treatment.R1.fastq.gz -> /CRISPR/BulkCrispr/00.RawData/Treatment.R1.fastq.gz
lrwxrwxrwx 1 crispr001 crispr001 51 Aug 14 11:14 Treatment.R2.fastq.gz -> /CRISPR/BulkCrispr/00.RawData/Treatment.R2.fastq.gz
```

3.3 数据质控

数据QC

```
mkdir -p ${WorkDir}/01.Fastqc  
cd ${WorkDir}/01.Fastqc
```

三个样本循环

for Sample in Baseline Control Treatment
do

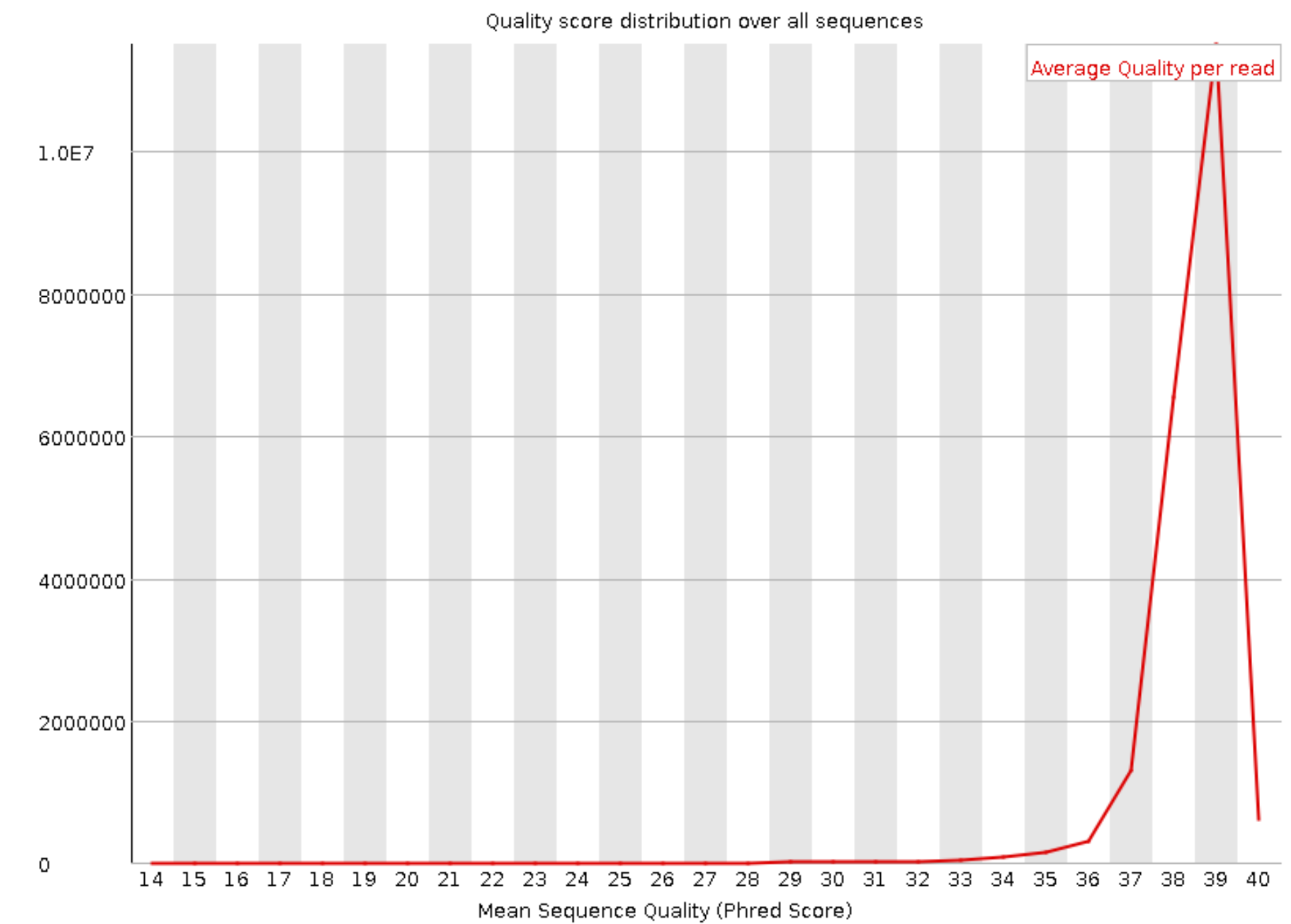
```
fastqc -t 3 --outdir ${WorkDir}/01.Fastqc \  
${WorkDir}/00.RawData/${Sample}.R1.fastq.gz  
done
```

```
cd ${WorkDir}
```

Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ✗ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ✓ [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ✗ [Overrepresented sequences](#)
- ✗ [Adapter Content](#)
- ✗ [Kmer Content](#)

✓ Per sequence quality scores

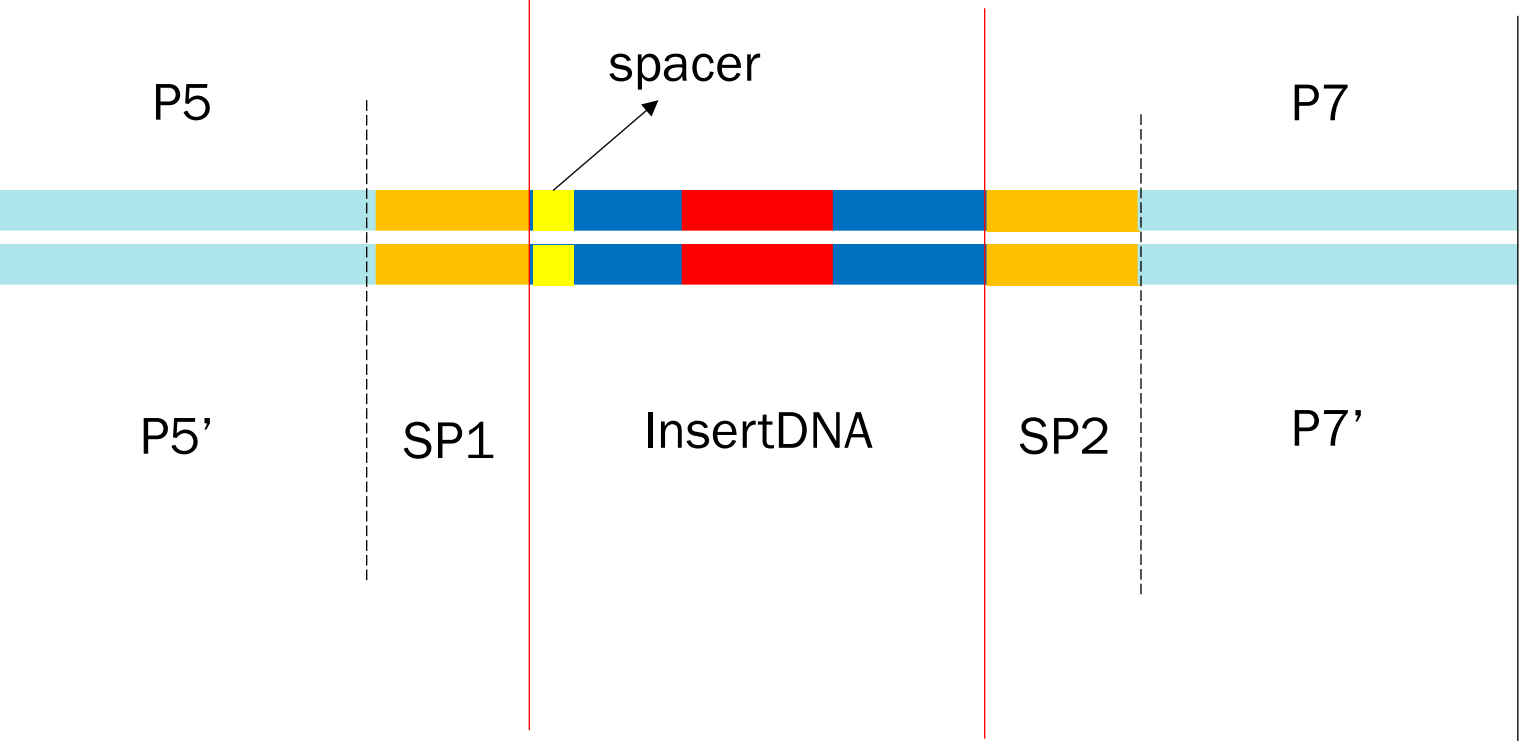


fastqc 不会对数据做任何改变仅作为QC参考

3.4.1 sgRNA 提取 --- 为什么提取？



上机文库结构



(Optional) Step 3: determine the trimming length and sgRNA length

****Note:** since version 0.5.6, MAGeCK is now able to automatically determine the trimming length and sgRNA length, in most cases. Therefore, you don't need to go to this step unless MAGeCK fails to do so by itself. ******

In many cases, your sequencing primer is not exactly in front of the first base of guide RNA. This is indeed the case in this demo, where the the first few bases in the fastq file are identical. Make sure you know exactly how many bases to trim before running MAGeCK. You can talk to experimental people, or get this information by taking a look at the first few lines of the fastq files.

Here are the first few lines of ERR376998.fastq (only sequences are shown):

```
CTTGTGGAAGGACGAAACACCGGTGAAGGTGCCGTTGTGTAGTTTTAGA
CTTGTGGAAGGACGAAACACCGAGCAGCACAACAATATGGGTTTTAGAG
CTTGTGGAAGGACGAAACACCGCTCTTGGGTTTGGATGTTTGTTTTAGA
CTTGTGGAAGGACGAAACACCGTTTGCGAGGGGAGCGCCGTTTTAGA
```

长度不同的spacer序列的引入使得下机数据 “不齐平”

```
5GAAAGGACGAAACACCGTCCCCGATGGCGCAGGTACCGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGTTTAGGGTCACACCCACCAAGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGGGGCTTCCACGTTCCCTATCGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGAGATGGCGGCCATGCGAATCGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGTATTCCACCACCAGATGCCAGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGCGGGCGCTGTTACGCTGCGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGACCGACTGGTCCCCGTCGTAGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGTAGGCATCCCGCACGCTCACGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGGATGTAGCAGGGTGTCTTGTGTTTTAGAGCTAGAAATAGCAAGTTA
TGTGGAAGGACGAAACACCGACCGCCATGACCAGCTCAAAGTTTTAGAGCTAGAAATAGCAAG
5GAAAGGACGAAACACCGCAAACCTCGTCCCTTCCATAGGTTTTAGAGCTAGAAATAGCAAGTTA
CTTGTGGAAGGACGAAACACCGGAATTCTGGTTATATCCTTCGTTTTAGAGCTAGAAATAGCA
5GAAAGGACGAAACACCGCAGCACCAGTTTGATGACGGTTTTAGAGCTAGAAATAGCAAGTTA
5GAAAGGACGAAACACCGTGAGACGAGCCAGAAGTACAGTTTTAGAGCTAGAAATAGCAAGTTA
TGTGGAAGGACGAAACACCGGCCGCCACCACTAGATGAGTTTTAGAGCTAGAAATAGCAAG
5GAAAGGACGAAACACCGAGAAGCATTGGGCCACCTTGTTTTAGAGCTAGAAATAGCAAGTTA
```

MAGeCK软件自动trim时要求序列齐平，
因此我们需要将sgRNA提取出来以保证 “齐平”

3.4.2 sgRNA 提取 --- 怎么做提取

sgRNA提取

```
mkdir -p ${WorkDir}/02.CleanData  
cd ${WorkDir}/02.CleanData
```

```
for Sample in Baseline Control Treatment  
do
```

```
cutadapt -g CACCG...GTTTT \
```

```
-m 20 -M 20 -q 20 \
```

```
--max-n 0 -j 3 \
```

```
--discard-untrimmed \
```

```
${WorkDir}/00.RawData/${Sample}.R1.fastq.gz \
```

```
--report=full \
```

```
-o ${WorkDir}/02.CleanData/${Sample}.sgRNA.fq
```

```
done
```

```
cd ${WorkDir}
```

```
(base) crispr001@izbp18758pdxsp0rsgmsuhZ:~/BulkCrispr$ ls -l 02.CleanData/  
total 1212944  
-rw-rw-r-- 1 crispr001 crispr001 417373876 Aug 14 13:56 Baseline.sgRNA.fq  
-rw-rw-r-- 1 crispr001 crispr001 411536576 Aug 14 13:57 Control.sgRNA.fq  
-rw-rw-r-- 1 crispr001 crispr001 413129079 Aug 14 13:57 Treatment.sgRNA.fq
```

利用sgRNA上下游的固定序列进行提取

3.5.1 sgRNA 比对 --- 所需文件

```
HGLibA_00001,GTCGCTGAGCTCCGATTCTGA,A1BG
HGLibA_00002,ACCTGTAGTTGCCGGCGTGC,A1BG
HGLibA_00003,CGTCAGCGTCACATTGGCCA,A1BG
HGLibA_00004,CGCGCACTGGTCCAGCGCAC,A1CF
HGLibA_00005,CCAAGCTATATCCTGTGCGC,A1CF
HGLibA_00006,AAGTTGCTTGATTGCATTCT,A1CF
HGLibA_00007,CGCTTCTTAAATTCTTGGGT,A2M
HGLibA_00008,TCACAGCGAAGGCGACACAG,A2M
HGLibA_00009,CAAACCTCTTCATCCAAGTC,A2M
HGLibA_00010,AAATTTCCCTCCGTTTCAGA,A2ML1
HGLibA_00011,GCTTACCTGTAGTTCCACCA,A2ML1
HGLibA_00012,TATTTCCAACCTCCGACACC,A2ML1
HGLibA_00013,CGAATGCGCGTCGCGTTCGA,A3GALT2
HGLibA_00014,CTTCGAACGCGACGCGCATT,A3GALT2
HGLibA_00015,CGGCAGATCCTACTTACACT,A3GALT2
HGLibA_00016,CAGTCCCGCTACGTCCTCAA,A4GALT
HGLibA_00017,CTACTGCCCCACGACGCACG,A4GALT
HGLibA_00018,CTTGAGGACATCAACCCCG,A4GALT
```

sgRNA库文件有三列（逗号分隔）：

sgRNA ID（唯一）

核酸序列（唯一）

靶向的基因名（可重复）

```
HGLibA_64384
HGLibA_64385
HGLibA_64386
HGLibA_64387
HGLibA_64388
HGLibA_64389
HGLibA_64390
HGLibA_64391
HGLibA_64392
HGLibA_64393
HGLibA_64394
HGLibA_64395
HGLibA_64396
HGLibA_64397
```

Control sgRNA文件有一列：

sgRNA ID（唯一且必须在库中）

sgRNA比对

```
mkdir -p ${WorkDir}/03.MAGeCK
```

```
cd ${WorkDir}/03.MAGeCK
```

```
mageck count --trim-5 0 \
```

```
-l /CRISPR/BulkCrispr/01.Lib/lib.csv \
```

```
--control-sgrna /CRISPR/BulkCrispr/01.Lib/control.txt \
```

```
--norm-method control -n ${WorkDir}/03.MAGeCK/BulkCrispr \
```

```
--sample-label Baseline,Control,Treatment \
```

```
--fastq ../02.CleanData/Baseline.sgRNA.fq ../02.CleanData/Control.sgRNA.fq ../02.CleanData/Treatment.sgRNA.fq
```

```
cd ${WorkDir}
```

3.5.2 sgRNA 比对 --- 结果



```
(base) crispr001@izbp18758pdxsp0rsgmsuhZ:~/BulkCrispr$ ls -l 03.MAGeCK/
total 11976
-rw-rw-r-- 1 crispr001 crispr001 8814796 Aug 14 14:42 BulkCrispr.count_normalized.txt
-rw-rw-r-- 1 crispr001 crispr001 8882 Aug 14 14:42 BulkCrispr.count_report.Rmd
-rw-rw-r-- 1 crispr001 crispr001 107 Aug 14 14:42 BulkCrispr_countsummary.R
-rw-rw-r-- 1 crispr001 crispr001 8540 Aug 14 14:42 BulkCrispr_countsummary.Rnw
-rw-rw-r-- 1 crispr001 crispr001 444 Aug 14 14:42 BulkCrispr.countsummary.txt
-rw-rw-r-- 1 crispr001 crispr001 3402883 Aug 14 14:42 BulkCrispr.count.txt
-rw-rw-r-- 1 crispr001 crispr001 4597 Aug 14 14:42 BulkCrispr.log
```

BulkCrispr.count.txt 用于下游分析

sgRNA	Gene	Baseline	Control	Treatment	
HGLibA_10276	CNGB1	20	40	32	
HGLibA_33826	OR2L5	47	33	34	
HGLibB_56764	ZNF83	35	43	39	
HGLibB_55594	ZNF248	140	121	93	
HGLibB_51486	TRIM56	81	93	70	
HGLibA_12580	DCX	27	25	40	
HGLibB_12152	CYP3A7-CYP3AP1	13	23	23	17
HGLibB_18402	GADL1	37	26	36	
HGLibA_52472	TYROBP	50	34	20	
HGLibB_31228	NCAPG	29	1	1	
HGLibA_03864	AUH	19	7	0	
HGLibA_32921	NUDCD1	18	23	15	
HGLibA_49416	THUMPD1	15	9	7	
HGLibA_46887	SPTBN4	55	57	51	
HGLibA_57098	hsa-let-7a-1	66	63	63	102
HGLibA_34747	OTOS	40	72	22	
HGLibB_49525	TKT	43	85	49	
HGLibB_21581	HLA-DOA	25	13	16	
HGLibB_13743	DOHH	27	39	63	
HGLibA_50285	TMEM240	71	58	56	
HGLibB_45880	SNPH	64	68	60	
HGLibB_08992	CELA3A	32	26	40	
HGLibB_43150	SEC14L5	22	10	7	
HGLibB_07929	CCDC96	46	51	16	

BulkCrispr.countsummary.txt 用于查看比对情况

File	Label	Reads	Mapped	Percentage	TotalsgRNAs	Zerocounts	GiniIndex	NegSelQC
../02.CleanData/Baseline.sgRNA.fq	Baseline	3847930	3486229	0.906	119461	2118	0.1645	0
../02.CleanData/Control.sgRNA.fq	Control	3759450	3358394	0.8933	119461	4477	0.1999	0
../02.CleanData/Treatment.sgRNA.fq	Treatment	3774021	3377413	0.8949	119461	6027	0.2173	0

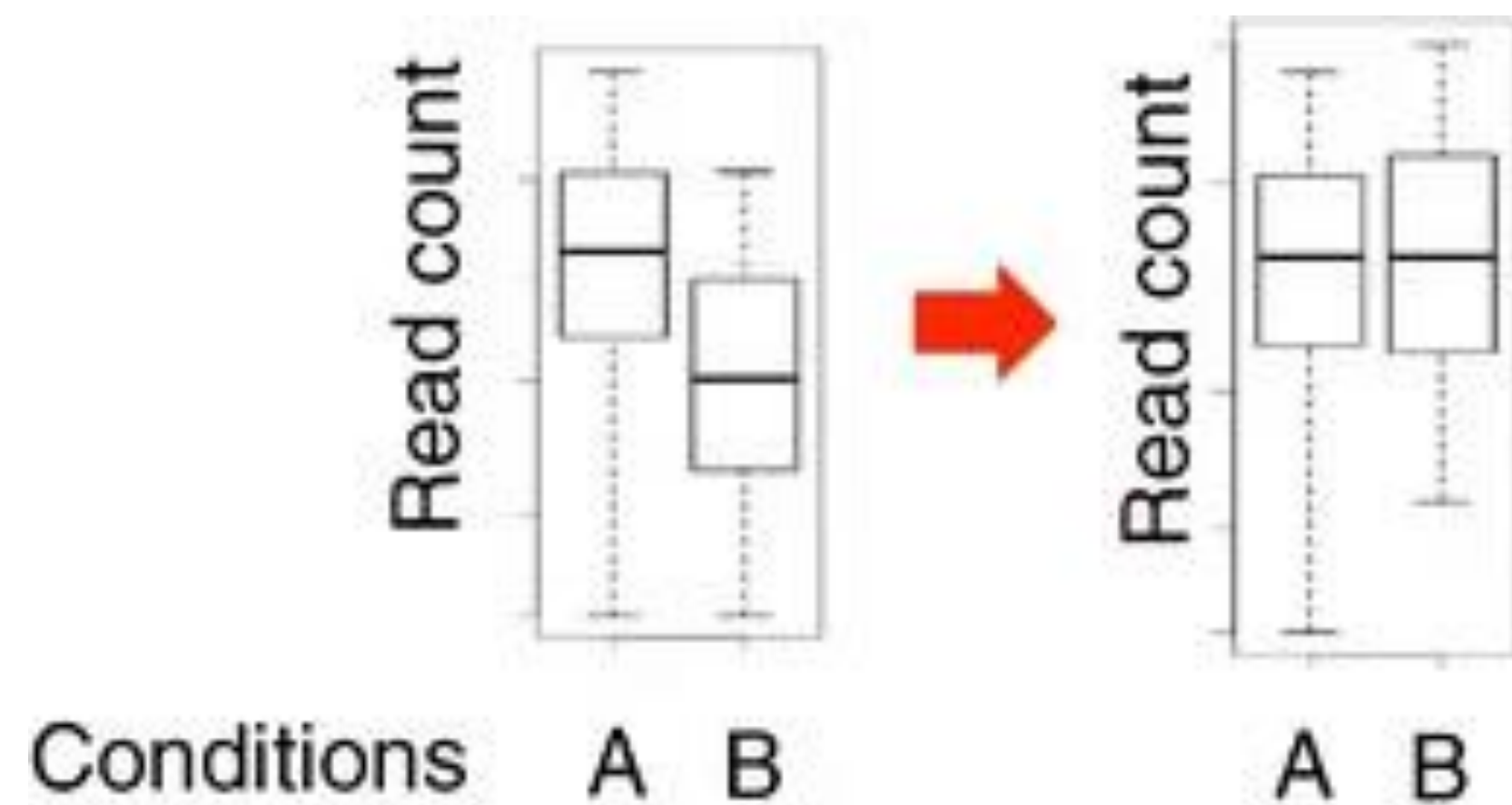
3.6.1 差异基因筛选 --- 差异算法的选则

sgRNA	Gene	Baseline (Day0)	Control (Day7)	Treatment (Day14)	
sgRNA1	gene1	10	20	30	趋于正筛
sgRNA2	gene1	10	20	29	
sgRNA3	gene1	10	15	22	
sgRNA4	gene2	10	5	0	趋于负筛
sgRNA5	gene2	10	4	1	
sgRNA6	gene2	10	6	0	

α -RRA : Treatment – Control

MLE : (Treatment – Baseline) – (Control – Baseline)

3.6.2.1 α -RRA 之数据标准化



对于 sgRNA i 在 实验 j 中的 count 数记为 k_{ij}
($1 \leq i \leq M, 1 \leq j \leq N$, M 为 sgRNA 数, N 为实验样本数)

单个 sgRNA 的 size factor $\hat{s}_j$ 为实验 j 与 “平均样本” 的 count 比的中位数即:

$$\hat{s}_j = \text{median}_i \frac{k_{ij}}{\left(\prod_{v=1}^m k_{iv} \right)^{1/m}}$$

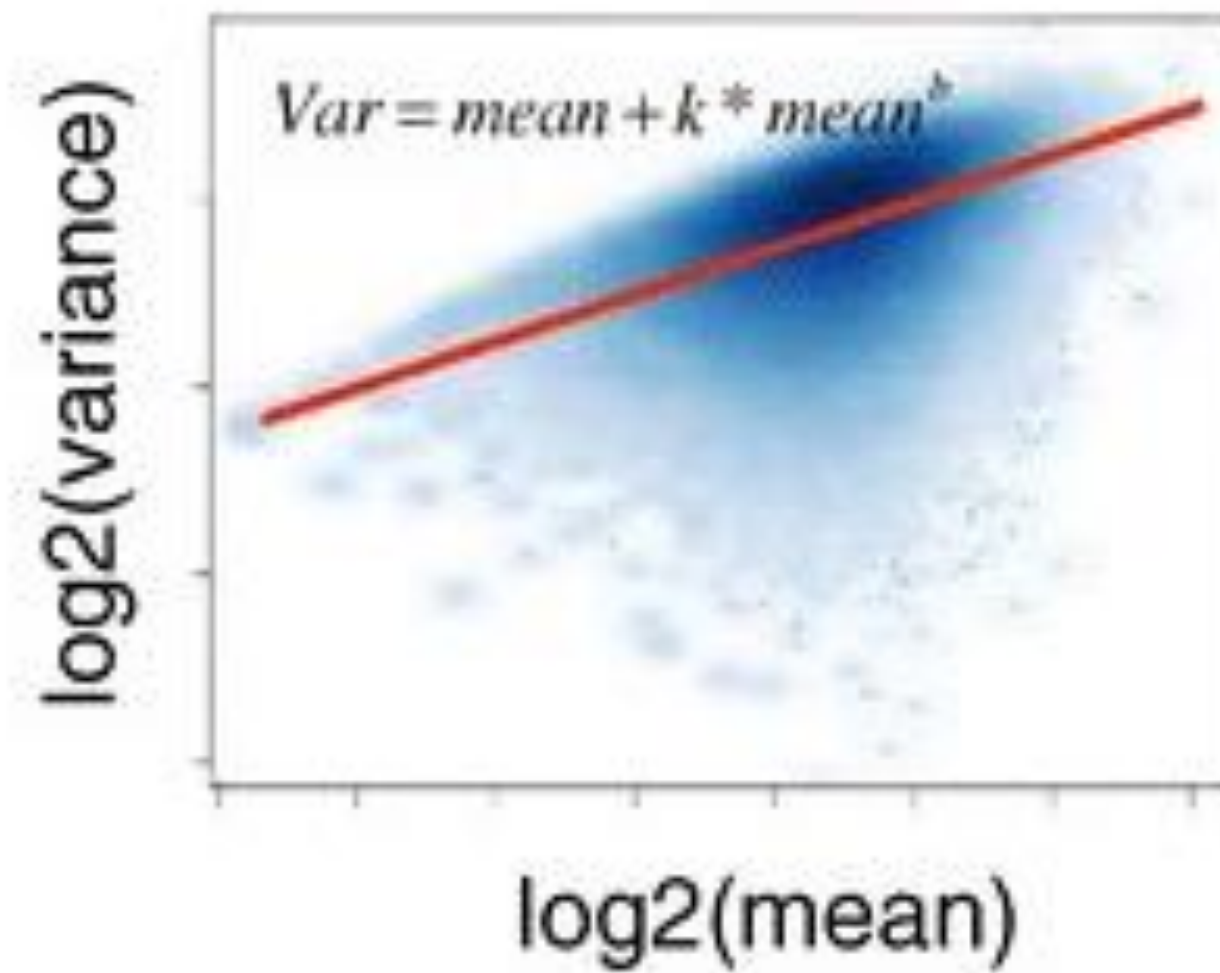
Normalized count = absolute count / size factor

标准化是采用 绝对定量 count 数 / 一个固定值
最终样本内的各 sgRNA 数据比值不变

其中 s_j 为实验 j 中所有 sgRNA size factor 的中位值

Normalized count $k'_{ij} = k_{ij} / s_j$

3.6.2.2 α -RRA 之建模矫正



Mean-variance modeling

理想状态下数据符合泊松分布

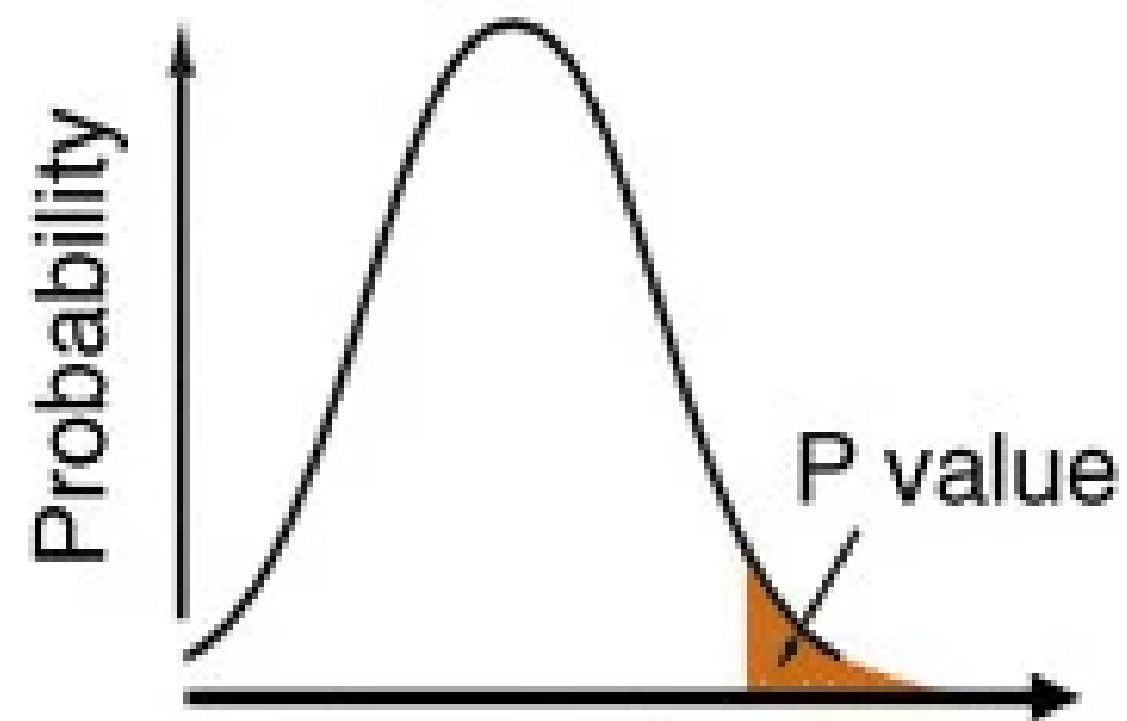
实际测序中，有些sgRNA丰度升高，有些sgRNA丰度降低
数据往往趋于离散 ($\hat{\sigma}^2 \gg \hat{\mu}$)

假定Variance是与mean相关的平滑函数且满足以下等式

$$\hat{\sigma}^2 = \hat{\mu} + k\hat{\mu}^b \quad k \geq 0, b \geq 0$$

通过建模重新计算方差补偿数据过度离散

3.6.2.3 α -RRA 之计算sgRNA的双尾P值



sgRNA test and ranking

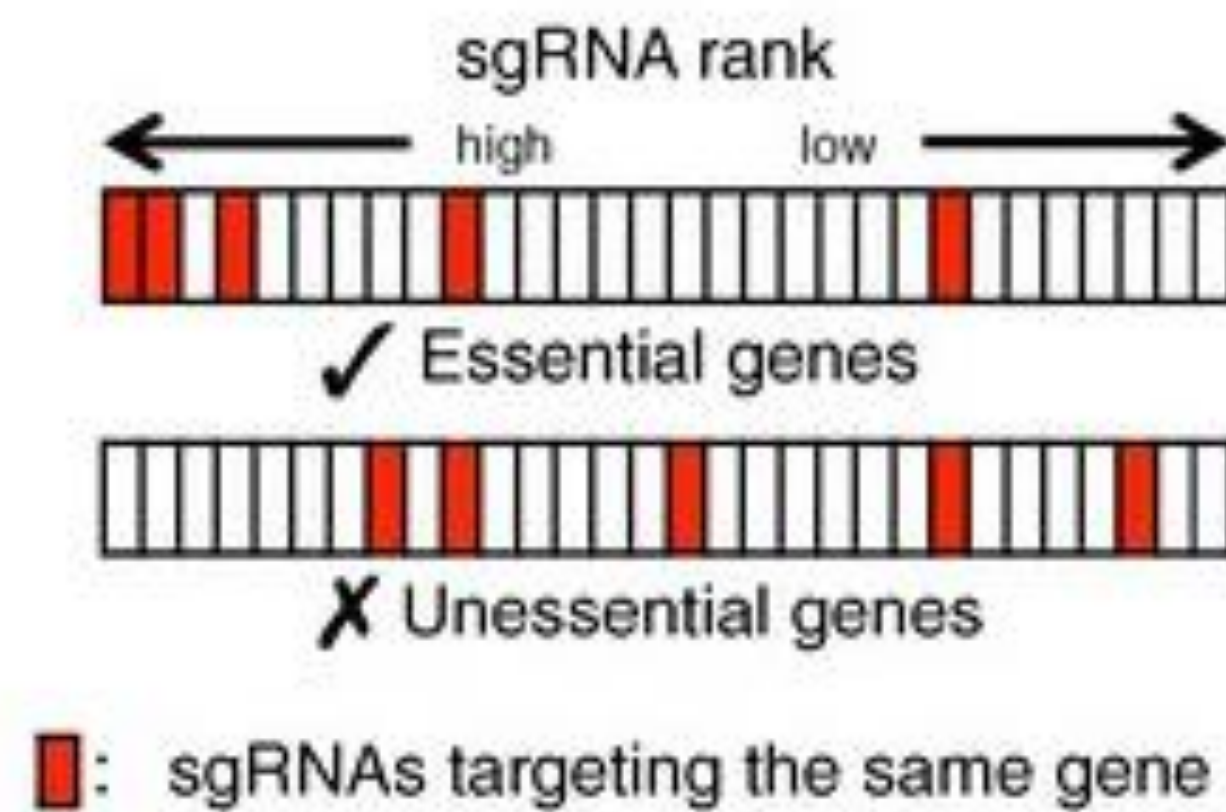
假定sgRNA i 在condition A中的read count为 x_{iA} 符合负二项分布则有

$x_{iA} \sim NB(\mu_{iA}, \sigma_{iA}^2)$ x_{iA} 为normalized count σ_{iA}^2 也是模型校正后的方差

根据PMF得到两个单尾P值 正筛 $p_{\text{high}}(\mu_{iB} > \mu_{iA})$ 和负筛 $p_{\text{low}}(\mu_{iB} < \mu_{iA})$

$$p = \begin{cases} \sum_{x > \mu_{iB}} NB(x | \mu_{iA}, \sigma_{iA}^2), & \mu_{iB} > \mu_{iA} \\ \sum_{x < \mu_{iB}} NB(x | \mu_{iA}, \sigma_{iA}^2), & \mu_{iB} < \mu_{iA} \end{cases}$$

3.6.2.4 α -RRA 之挑选差异基因



Gene test and ranking

将sgRNA的双尾P值分别转化为排名列表

靶向一个基因的多数sgRNA排名都靠前，那么这个基因就是重要基因，反之则不是。

经典RRA算法需要假设各sgRNA在rank list中均匀分布，但实际情况中这一假设未必成立，因此作者对排名中的NB P-value设置了阈值。

3.6.2.5 α -RRA 之代码实现



基因筛选 (α -RRA算法)

```
mkdir -p ${WorkDir}/04.RRA
cd ${WorkDir}/04.RRA

mageck test \
-k ${WorkDir}/03.MAGeCK/BulkCrispr.count.txt \
-t Treatment \
-c Control \
-n ${WorkDir}/04.RRA/RRA \
--norm-method control \
--control-sgrna /CRISPR/BulkCrispr/01.Lib/control.txt

cd ${WorkDir}
```

RRA算法会提供两组排名
正筛与负筛排名
同时会提供对应的 P 值与 FDR 值

id	num	neg score	neg p-value	neg fdr	neg rank	neg goodsgrna	neg lfc	pos score	pos p-value	pos fdr	pos rank	pos goodsgrna	pos lfc
EME1	6	3.6008e-06	3.4037e-05	0.737624	1	5	-1.747	0.85272	0.83889	1.0	17787	1	-1.747
XRCC6	5	1.0351e-05	8.2466e-05	0.740099	2	5	-2.4328	0.99999	0.99999	1.0	20686	0	-2.4328
CNOT6L	6	2.1071e-05	0.00015283	0.740099	3	3	-0.69988	0.40339	0.52546	0.93196	12046	2	-0.69988
PKLR	6	2.5582e-05	0.00017841	0.740099	4	3	-0.33962	0.65731	0.74638	1.0	15777	1	-0.33962
AP2A2	6	3.1635e-05	0.00021359	0.740099	5	3	-0.68145	0.30324	0.43303	0.901607	10250	2	-0.68145
RCSD1	5	4.7818e-05	0.00020354	0.740099	6	4	-2.3854	0.97296	0.97037	1.0	20088	0	-2.3854
TCEB1	6	5.6322e-05	0.00028943	0.740099	7	6	-2.1389	0.99994	0.99996	1.0	20685	0	-2.1389
ADAMTS1	6	6.9691e-05	0.00032278	0.740099	8	4	-0.44754	0.23261	0.33659	0.837672	8585	2	-0.44754
PRG4	6	7.571e-05	0.00034152	0.740099	9	3	-0.87967	0.75645	0.78768	1.0	16705	1	-0.87967
NRIP2	6	7.6744e-05	0.00034152	0.740099	10	2	1.0035	0.038114	0.060038	0.54161	2351	4	1.0035
MEGF11	6	0.00011255	0.00056721	0.779121	11	5	-1.9412	0.67447	0.7584	1.0	16038	1	-1.9412
TAOK3	6	0.0001279	0.00063986	0.779121	12	2	0.1035	0.14948	0.22168	0.718402	6573	3	0.1035
FAM5B	4	0.00013438	0.00040822	0.779121	13	4	-2.4734	0.99987	0.99988	1.0	20684	0	-2.4734
GPKOW	6	0.00013489	0.00064305	0.779121	14	6	-1.0244	0.99987	0.9999	1.0	20683	0	-1.0244
IFI44L	6	0.00013735	0.00064534	0.779121	15	5	-0.88857	0.78974	0.80448	1.0	17052	1	-0.88857
PRPF19	6	0.00013828	0.00064534	0.779121	16	4	-2.1952	0.8234	0.81511	1.0	17429	1	-2.1952
FBXL20	6	0.0001424	0.00067778	0.779121	17	3	-0.69399	0.84402	0.83003	1.0	17664	1	-0.69399
TRAP1	6	0.00015792	0.00071022	0.779121	18	6	-1.1821	0.99984	0.99989	1.0	20682	0	-1.1821
DNAJC4	6	0.00015823	0.00072164	0.779121	19	6	-1.8222	0.9998	0.99987	1.0	20680	0	-1.8222
FBX025	6	0.00017906	0.00077601	0.779121	20	4	-1.1286	0.97937	0.97981	1.0	20229	0	-1.1286

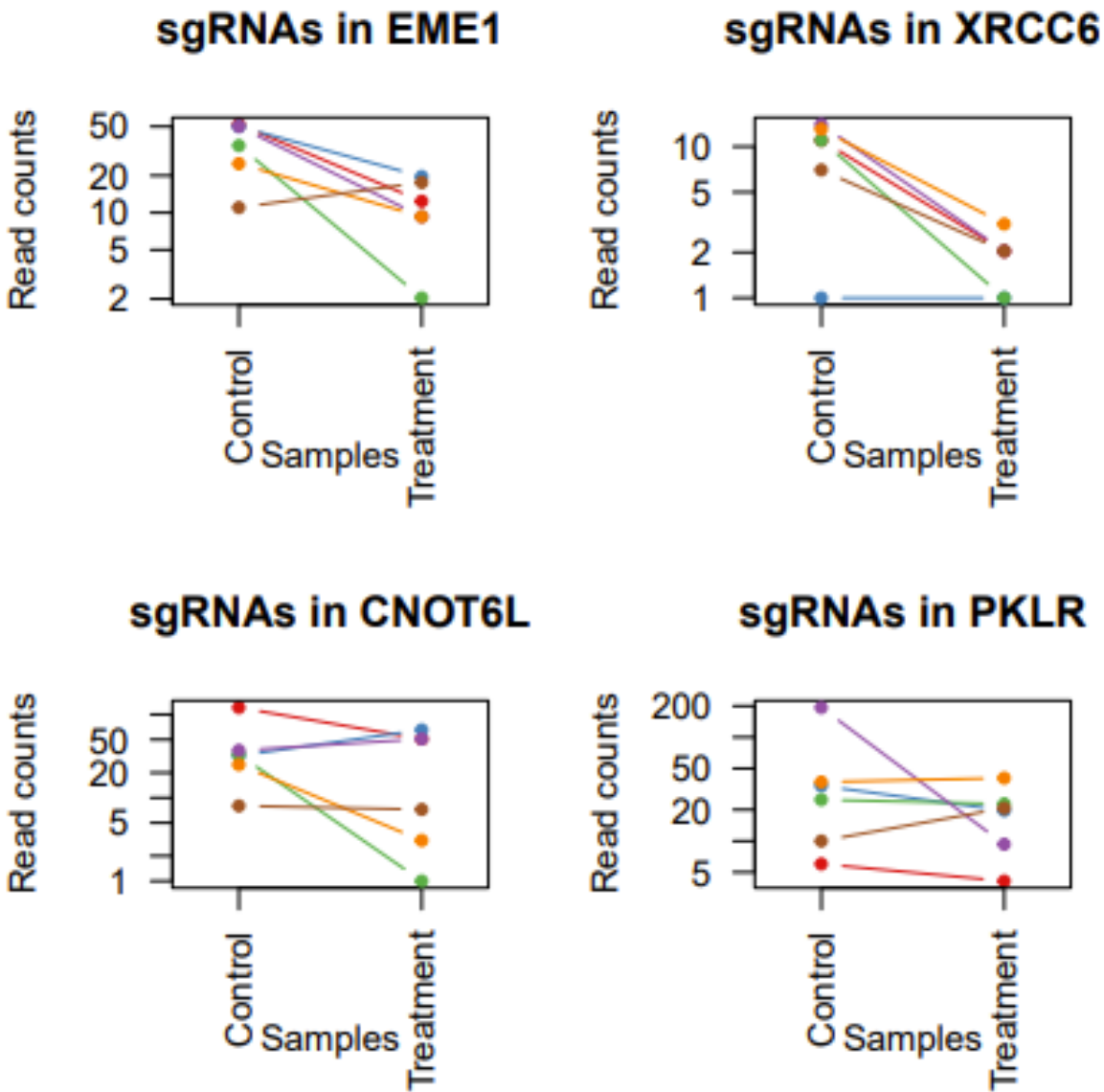
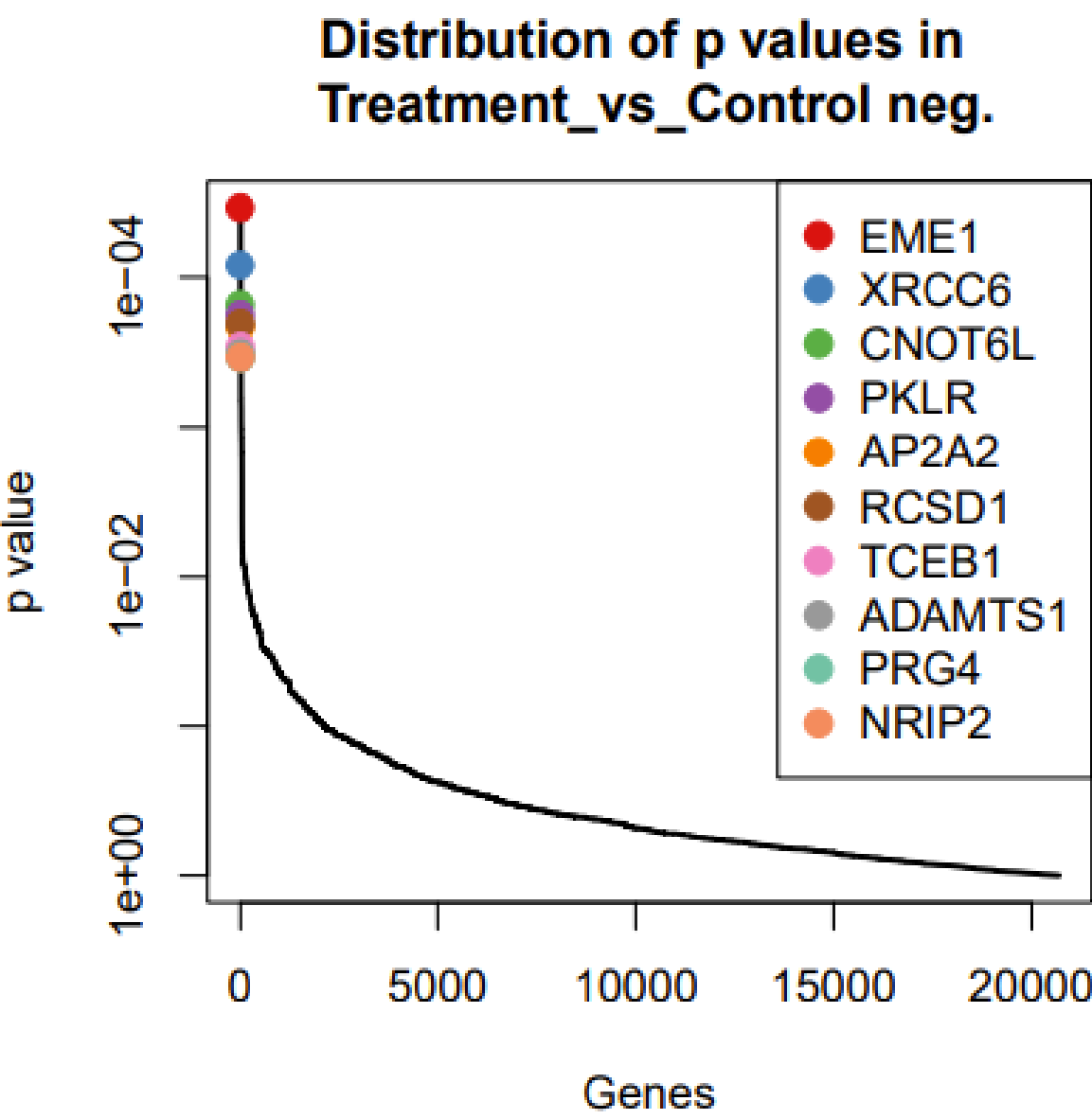
3.6.2.6 α-RRA 之筛选结果处理



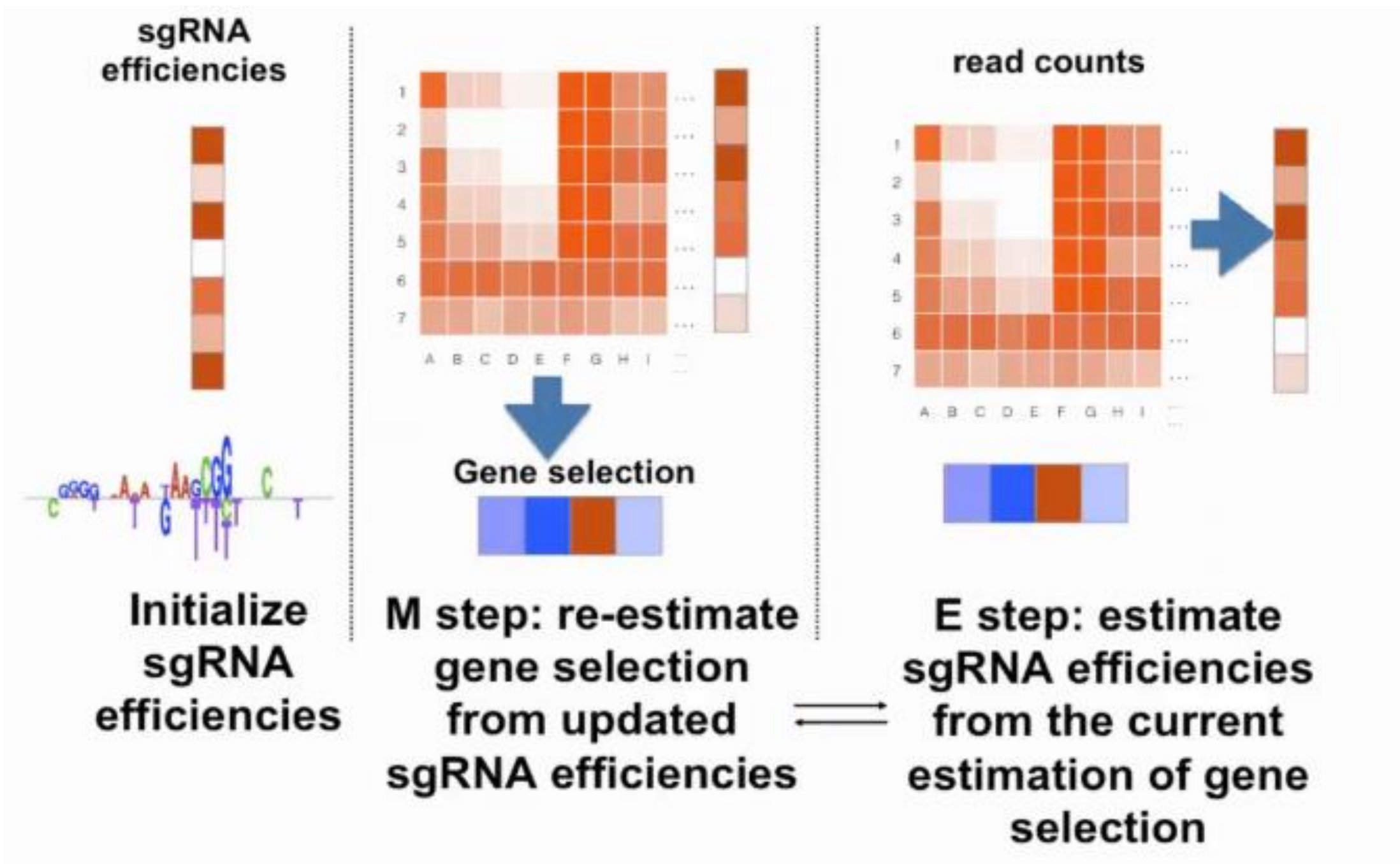
筛选结果处理 (α-RRA算法)

```
cd ${WorkDir}/04.RRA
awk -F "\t" '{for(i=1;i<8;i++){printf $i "\t"};print $8}' \
RRA.gene_summary.txt|sort -k 6 -n > RRA.negative.txt
awk -F "\t" '{printf $1 "\t" $2 "\t";for(i=9;i<14;i++){printf $i "\t"};print $14}' \
RRA.gene_summary.txt|sort -k 6 -n > RRA.positive.txt
Rscript RRA.R
cd ${WorkDir}
```

id	num	pos score	pos p-value	pos fdr	pos rank	pos goodsgrna	pos lfc
GTSF1L	6	2.3755e-07	2.2844e-07	0.00495	1	5	2.0803
WNT6	6	7.4141e-06	4.7972e-06	0.034653		2	1.3195
LEMD1	6	1.414e-05	1.028e-05	0.050495		3	1.6196
PAX9	6	2.5582e-05	1.7133e-05	0.061881		4	0.29862
hsa-mir-1268b	4	4.4563e-05	1.165e-05	0.050495		5	1.9582
ARL4A	6	4.5848e-05	4.0891e-05	0.124381		6	0.7576
PLEKHA6	6	5.5145e-05	5.5511e-05	0.133663		7	1.7728
MUC20	5	6.862e-05	4.5916e-05	0.124381		8	0.62034
ARHGEF19	6	7.4964e-05	7.7898e-05	0.142001		9	1.3453
GATA4	6	7.6744e-05	7.8354e-05	0.142001		10	0.26342
FAM160B1	6	0.00010275	0.00010348	0.142001		11	0.86252
FAM174B	6	0.00010275	0.00010348	0.142001		12	-0.13
CCNK	5	0.00010516	0.00010622	0.142001		13	1.6352
ARL16	6	0.00011024	0.00011262	0.142001		14	1.6453
PPP1R21	6	0.00011377	0.00011856	0.142001		15	1.3796
RPS19	6	0.00011377	0.00011856	0.142001		16	1.5969
ZP2	6	0.0001202	0.00011902	0.142001		17	1.134
ADIPOQ	6	0.0001279	0.0001245	0.142001		18	0.042343
IQCJ-SCHIP1	2	0.00013018	2.9697e-06	0.032178		19	2.4585
CAMK1	6	0.00014194	0.00016105	0.145406		20	1.6404
SHC4	6	0.00014874	0.00016333	0.145406		21	1.3484
LOC100129924	6	0.00015295	0.00017658	0.145406		22	0.81149
KLHL10	6	0.00016068	0.0001807	0.145406		23	-0.30304
PCDHB16	6	0.00016399	0.0001807	0.145406		24	0.98924
LAT	5	0.00017697	0.00016516	0.145406		25	0.93512
CARNS1	6	0.00017906	0.00019897	0.145406		26	0.80429



3.6.3.1 MLE 算法概览



对于每个sgRNA，MLE算法的目标是估计它在处理组中的富集程度。
即目标是找到使对数似然函数 $\ell(\theta)$ 最大化的参数值 θ

E步会计算给定当前参数估计值 $\theta^{(t)}$ 的条件期望，然后在M步中更新参数 θ 以最大化这个期望。

当达到收敛条件 $|\ell(\theta^{(t+1)}) - \ell(\theta^{(t)})| < \epsilon$
或迭代次数达到预设的上限时，迭代停止，输出估计值

MAGeCK会将所有与该基因相关的sgRNA的 θ_i 值结合起来，
通过加权平均等方式得到该基因的一个综合富集评分。

该值类似于 LFC 。

3.6.3.2 MLE 算法的实现



基因筛选 (MLE算法)

```
mkdir -p ${WorkDir}/05.MLE
cd ${WorkDir}/05.MLE
mageck mle -k
${WorkDir}/03.MAGeCK/BulkCrispr.count.
txt \
-d /CRISPR/BulkCrispr/01.Lib/matrix \
-n ${WorkDir}/05.MLE/MLE \
--norm-method control \
--control-sgrna
/CRISPR/BulkCrispr/01.Lib/control.txt \
--threads 3
cd ${WorkDir}
```

Gene	sgRNA	Control beta	Control z	Control p-value	Control fdr	Control wald-p-value
CNGB1	6	0.2474 0.84931	0.067751	0.12241 0.39571	0.6469 0.25594	0.87814 0.10094 0.1895
OR2L5	6	-0.2464 -1.2517	0.059686	0.11022 0.21068	0.50133 -0.42921	-2.1738 0.0029
ZNF83	6	-0.070773	-0.27745	0.57791 0.67093	0.78144 0.88681	-0.048519 -0.190
ZNF248	6	-0.34285	-1.2598 0.0099092	0.026001	0.20774 0.49859	-0.31277
TRIM56	6	-0.26502	-1.5305 0.043693	0.086143	0.12589 0.41589	-0.52038
DCX	6	-0.27726	-1.2551 0.035028	0.071929	0.20946 0.5	-0.25476
CYP3A7-CYP3AP1	2	0.19486	0.49522 0.41061 0.51634	0.62044 0.79229	0.40946 1.0421	0.12873 0.2281
GADL1	6	-0.35235	-2.1527 0.0082961	0.022624	0.03134 0.28235	-0.23676
TYROBP	6	-0.095371	-0.28075	0.45744 0.56042	0.7789 0.88505	-0.52721 -1.538
NCAPG	6	-1.0549 -2.4299	0	0.015103	0.25142 -1.2084	-2.7768 0 0
AUH	6	-0.26038	-0.69473	0.046827	0.090658	0.48722 0.70787 -0.419
NUDCD1	6	-0.17697	-0.697 0.17625 0.26602	0.4858 0.70689	-0.16194	-0.63746
THUMPD1	6	-0.29366	-1.1363 0.02558 0.055942	0.25581 0.54191	-0.14767	-0.572
SPTBN4	6	-0.23717	-1.0801 0.070102	0.12599 0.28008	0.5584 -0.34416	-1.563
hsa-let-7a-1	4	-0.11868	-0.6436 0.4526	0.55596 0.51983	0.72896 0.088501	0.4807
OTOS	6	-0.39964	-1.455 0.0026732	0.0088388	0.14566 0.43664	-0.94023
TKT	6	-0.73691	-2.109 0	0.034949	0.28658 -0.69396	-1.986
HLA-D0A	6	-0.18409	-0.66477	0.15832 0.24334	0.50619 0.71976	-0.1764 -0.63652
DOHH	6	-0.17928	-0.48725	0.16965 0.2575	0.62608 0.79625	0.016554 0.0450
TMEM240	6	-0.53559	-2.4661 4.6089e-05	0.00027255	0.01366 0.24851	-0.29238
SNPH	6	-0.38486	-1.5873 0.0038715	0.012031	0.11245 0.40037	-0.28258
CELA3A	6	-0.42902	-2.2343 0.0013366	0.0049751	0.025462	0.26883 0.0151
SEC14L5	6	-0.77135	-2.132 0	0.033008	0.28393 -0.87665	-2.415
CCDC96	6	-0.048219	-0.19577	0.7053 0.77981	0.84479 0.92185	-0.47351 -1.882
CCDC69	6	-0.12102	-0.53366	0.3488 0.45574	0.59358 0.77643	0.20285 0.89845 0.2027
LETM5	6	-0.48907	-2.2843 0.00023045	0.0011033	0.022351	0.2633 -0.390

3.6.3.3 MLE 结果后续处理



MLE后续处理

cd \${WorkDir}/05.MLE

这一段为 R 代码，需先按 R + 回车进入R界面

```
library(MAGeCKFlute)
FluteMLE("MLE.gene_summary.txt",treatname="Treatment",
ctrlname="Control",proj="MLE",
organism="hsa",scale_cutoff=1)
q(save="no")
### R 代码至此结束
```

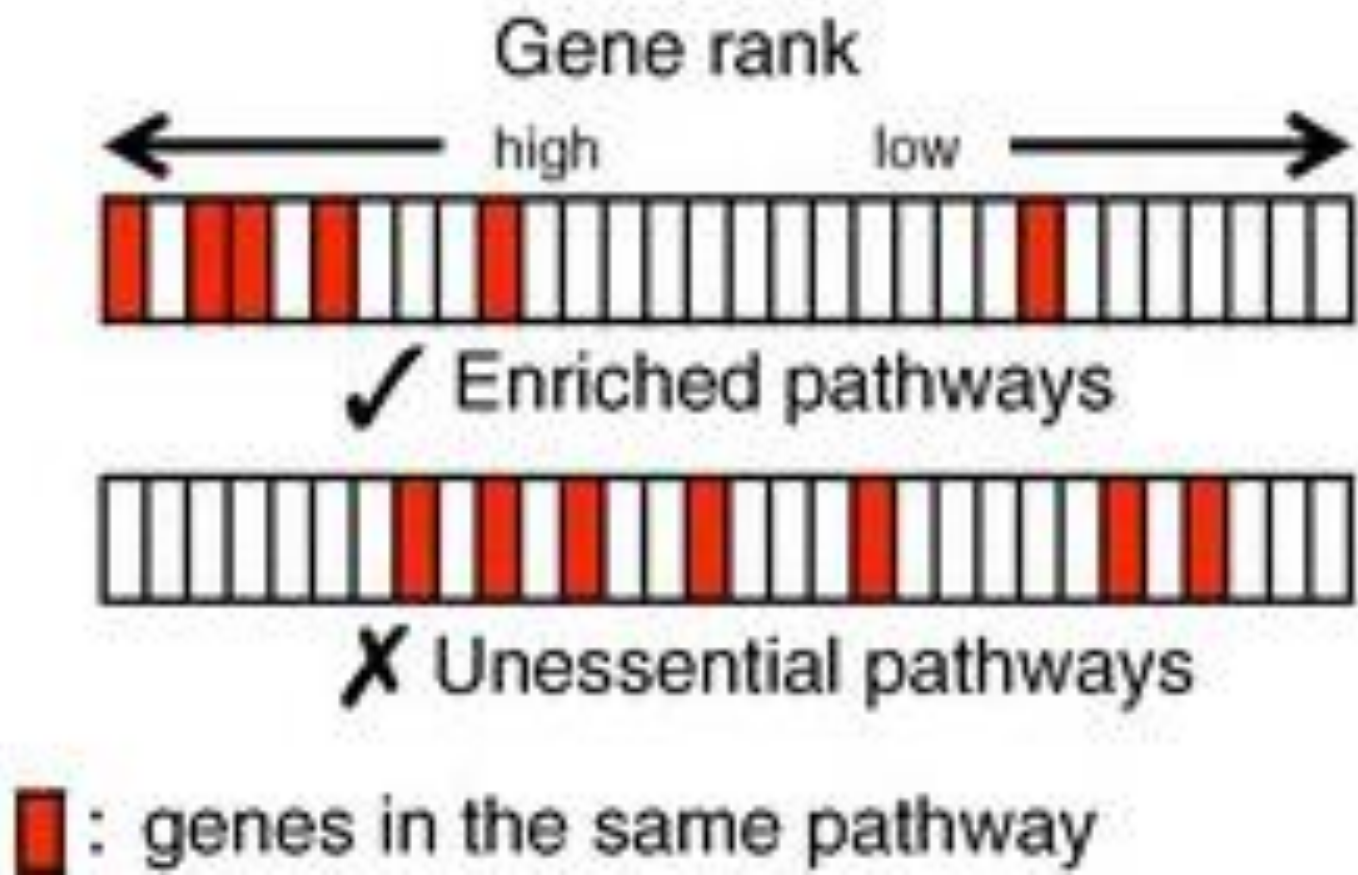
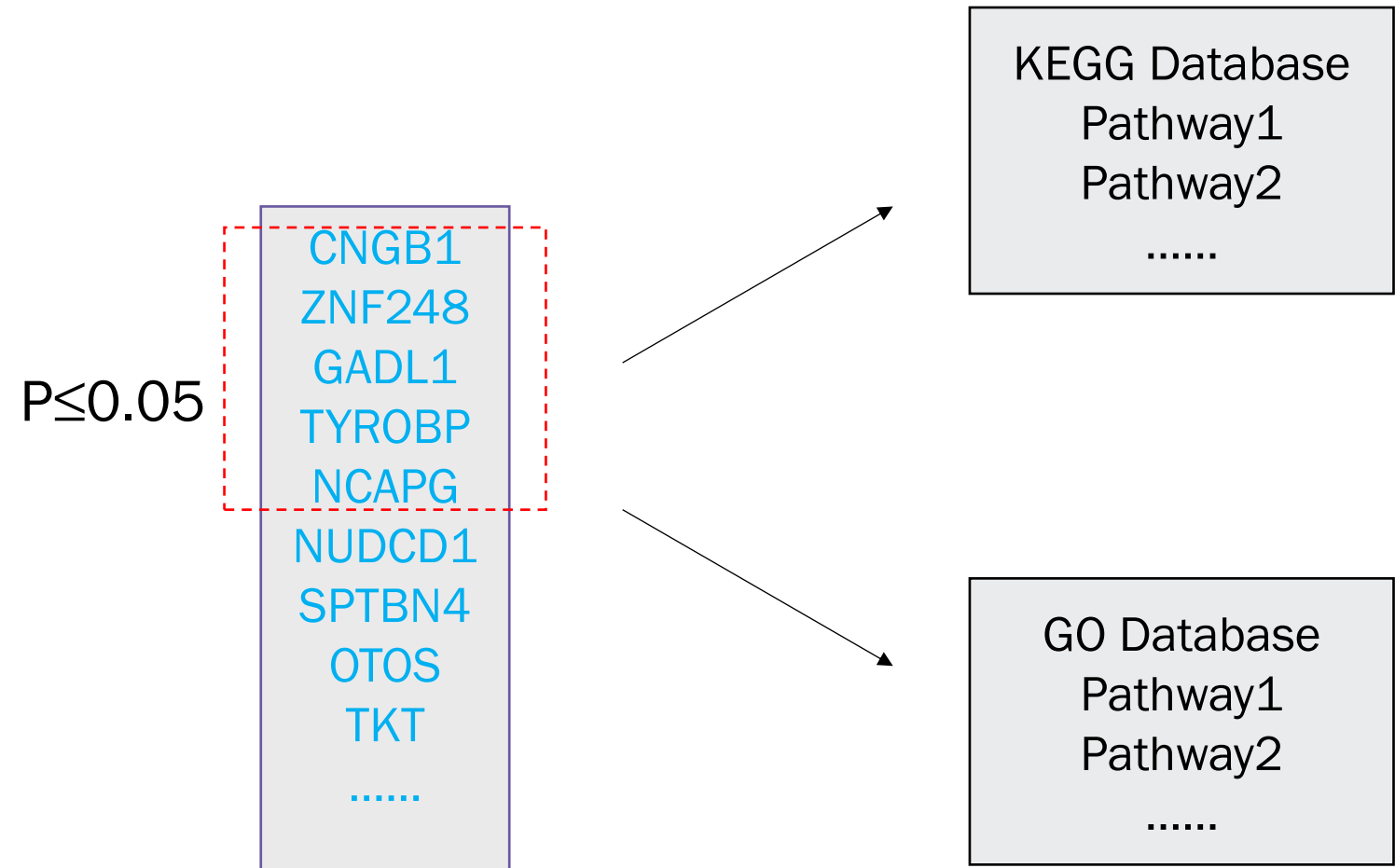
```
awk -F "\t" '{print $1 "\t" $2 "\t" $3 "\t" $5 "\t" $8 "\t" $9 "\t"
$12}' \
MAGeCKFlute_MLE/Selection/Data_ScatterView_TreatvsCtrl.
txt \
| sort -k 6 -n > MLE.selection.txt

cd ${WorkDir}
```

Gene	Control	Treatment	EntrezID	Symbol	HumanGene	GeneID	Diff	Rank	RandomIndex	Label
TMEM240	-0.569946367002937	-0.303720939895706	339453	TMEM240	TMEM240	339453	0.266225427107231	14918		
CELA3A	-0.456540246030733	0.0157303720939896	10136	CELA3A	CELA3A	10136	0.472270618124722	17579		
CCDC69	-0.128783041757119	0.21071821827021	26112	CCDC69	CCDC69	26112	0.339501260027329	16213		
CXXC1	-0.61999318946069	0.0562524671223485	30827	CXXC1	CXXC1	30827	0.676245656583039	18305		
SEPN1	-0.806240156642404	-0.506731348555045	57190	SEPN1	SEPN1	57190	0.299508808087359	15532		
IDH3A	-1.11554505597412	-0.761192944549478	3419	IDH3A	IDH3A	3419	0.354352111424643	16413		
FFAR4	-0.636178861788618	-0.197130866557248	338557	FFAR4	FFAR4	338557	0.43904799523137	17324		
GALNT9	-0.353179670540161	-0.0577618266054474	50614	GALNT9	GALNT9	50614	0.295417843934713	15466		
SMPD2	-0.193887540969651	0.238661625080506	6610	SMPD2	SMPD2	6610	0.432549166050157	17271		
hsa-mir-346	-0.517569063125186	-0.152660336982943	442911	hsa-mir-346	hsa-mir-346	442911	0.36496			
AIF1L	-0.201783509981697	0.485675108553383	83543	AIF1L	AIF1L	83543	0.68745861853508	18326		
SYDE1	-0.781881837142979	-0.273492198699437	85360	SYDE1	SYDE1	85360	0.508389638443542	17799		
EMC8	-0.301557910867067	0.0272120998067854	10328	EMC8	EMC8	10328	0.328770010673852	16041		
SAMD1	-0.492082748052611	-0.105925248789812	90378	SAMD1	SAMD1	90378	0.3861574992628	16800	4767	
NOVA1	-0.186928021112672	0.107005588681362	4857	NOVA1	NOVA1	4857	0.293933609794034	15427		
FLOT2	-0.569201464265952	-0.279060104294351	2319	FLOT2	FLOT2	2319	0.2901413599716	15360	10495	
NRL	-0.33487634614566	0.0276567012236927	4901	NRL	NRL	4901	0.362533047369353	16525		
OR4F17	-1.57312816583663	-1.13903143373569	81099	OR4F17	OR4F17	81099	0.434096732100941	17282		
HPSE	-0.444887838930745	-0.143830635946232	10855	HPSE	HPSE	10855	0.301057202984513	15554		
KATNAL2	-0.272251308900524	0.253547462240043	83473	KATNAL2	KATNAL2	83473	0.525798771140567	17881		
BDP1	-0.557953432937471	-0.289115575592629	55814	BDP1	BDP1	55814	0.268837857344842	14976		

Gene	Control	Treatment	Symbol	Diff	Rank	group
PLN	1.55333503596816	-0.919940581305965	PLN	-2.47327561727413	1	bottom
hsa-mir-7-2	-0.75975822585451	-2.83537282114142	hsa-mir-7-2	-2.07561459528691	2	bottom
HIST2H4B	1.80298386753501	0.204412772941641	HIST2H4B	-1.59857109459337	3	bottom
hsa-mir-548g	0.901491933767505	-0.625599900276318	hsa-mir-548g	-1.52709183404382	4	bottom
SPDYE6	0.36824798876261	-1.1558598051233	SPDYE6	-1.52410779388591	5	bottom
TCEB1	-0.519761205465458	-1.9818004279808	TCEB1	-1.46203922251535	6	bottom
MAGEA9B	0.789160601030094	-0.667618889327488	MAGEA9B	-1.45677949035758	7	bottom
POC1B-GALNT4	-0.0488251819690972	-1.30336775185424	POC1B-GALNT4	-1.25454256988514	8	bottom
EME1	0.0459424083769634	-1.13165603639914	EME1	-1.17759844477611	9	bottom
RCSD1	-0.224247648235645	-1.36600668979702	RCSD1	-1.14175904156138	10	bottom
LY75-CD302	0.187406887157877	-0.942814700932832	LY75-CD302	-1.13022158809071	11	bottom
hsa-mir-7705	-0.154854637551611	-1.18910103255563	hsa-mir-7705	-1.03424639500402	12	bottom
C4B	-0.874462605882603	-1.89495772131386	C4B	-1.02049511543126	13	bottom
CYTH3	-0.483282254288511	-1.49367377890429	CYTH3	-1.01039152461577	14	bottom
DNAJC4	0.158376963350785	-0.844337564664575	DNAJC4	-1.00271452801536	15	bottom
APITD1-CORT	0.063262461158643	-0.931221822865809	APITD1-CORT	-0.994484284024452	16	bottom
hsa-mir-485	0.318701315285404	-0.656815490411984	hsa-mir-485	-0.975516805697388	17	bottom
KRTAP9-2	-0.0870887072745073	-1.05437018261899	KRTAP9-2	-0.967281475344486	18	bottom
NCAPG2	-0.432128719192951	-1.39810525003636	NCAPG2	-0.965976530843407	19	bottom
MIPOL1	0.138094325969438	-0.809839403319968	MIPOL1	-0.947933729289405	20	bottom
RNF4	0.0398735793640659	-0.907018054141649	RNF4	-0.946891633505715	21	bottom
hsa-mir-1299	-2.08242880858128	-3.026613757713	hsa-mir-1299	-0.944184949131724	22	bottom
hsa-mir-6785	-0.507278763887115	-1.44921363721355	hsa-mir-6785	-0.941934873326439	23	bottom

3.7.1 GSEA 与 KEGG、G O的区别



传统的GO/KEGG富集分析需要手动找差异基因，主观性大

GSEA是根据表达量变化进行排序的基因列表。GSEA分析的是排序的基因列表中某些基因集是否在排序的顶端或底端显著富集。

3.7.2 GSEA 的实现



GSEA

```
mkdir -p ${WorkDir}/06.GSEA
cd ${WorkDir}/06.GSEA
```

这一段为 R 代码，需先按 R + 回车进入 R 界面

```
library(MAGeCKFlute)
library(tidyr)
library(ggplot2)
gene_summary <- "~/BulkCrispr/04.RRA/RRA.gene_summary.txt"
Organism <- "hsa"
Outname <- "GSEA"
```

```
dd <- ReadRRA(gene_summary,score = c("lfc", "rra")[1])
universe = dd$id; geneList= dd$Score; names(geneList) = universe
```

#GO enrich using RRA

```
enrich_go = enrich.GSE(geneList = geneList,keytype = "Symbol", type =
"GOBP+GOCC+GOMF",
organism = Organism, pvalueCutoff = 1)
write.table(enrich_go,file=paste0(Outname,".GO.csv"),sep=";",row.names = F)
```

#KEGG enrich using RRA

```
enrich_kegg = enrich.GSE(geneList = geneList,keytype = "Symbol", type = "KEGG",
organism = Organism, pvalueCutoff = 1)
write.table(enrich_kegg,file=paste0(Outname,".KEGG_pathway.csv"),
sep=";",row.names = F)
q(save="no")
### R 代码至此结束
```

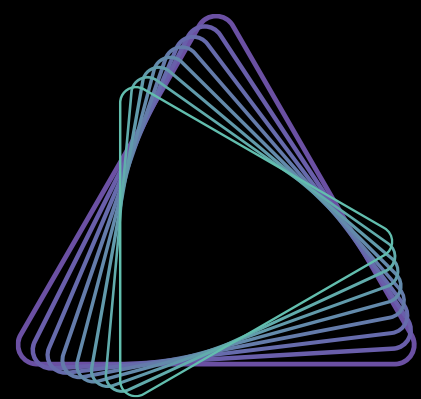
```
cd ${WorkDir}
```

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```
"ID","Description","NES","pvalue","p.adjust","geneID","geneName","Count"
"G0:0070979","protein K11-linked ubiquitination",-2.31699042332189,1.19576209358522e-06,0.012064
"G0:0051865","protein autoubiquitination",-2.13732295919192,2.42723554782755e-06,0.0122441897216
"G0:0005680","anaphase-promoting complex",-2.25329781969113,1.61502655166338e-05,0.0302369786626
"G0:0005697","telomerase holoenzyme complex",-2.25070517801031,1.69510873285502e-05,0.0302369786
"G0:0070122","isopeptidase activity",-2.13011435321056,1.73501804296365e-05,0.0302369786620262,"
"G0:0071005","U2-type precatalytic spliceosome",-2.10476019889735,1.79821460969528e-05,0.0302369
"G0:0006513","protein monoubiquitination",-2.18894400680673,2.1858822430382e-05,0.03150480850001
"G0:0031145","anaphase-promoting complex-dependent catabolic process",-2.13903552730014,3.270525
"G0:0071013","catalytic step 2 spliceosome",-1.93846439872601,3.8185882554107e-05,0.03995691116
"G0:0046540","U4/U6 x U5 tri-snRNP complex",-2.13488417732691,3.9604431725456e-05,0.039956911167
"G0:0000400","four-way junction DNA binding",-2.16241671840153,7.42768495154995e-05,0.0681253758
"G0:0042766","nucleosome mobilization",-1.78924346637858,8.41618575414669e-05,0.068673849858151,
"G0:0070936","protein K48-linked ubiquitination",-1.97517691959625,0.000100722767014734,0.068673
"G0:0035861","site of double-strand break",-1.94346070970037,0.000101287330811941,0.068673849858
"G0:0031463","Cul3-RING ubiquitin ligase complex",-2.07133619923828,0.000102102066396299,0.06867
"G0:0061578","Lys63-specific deubiquitinase activity",-2.03822168089333,0.000116075716614095,0.0
"G0:0070034","telomerase RNA binding",-2.10652293736073,0.000122691879824679,0.0728140220912461
"G0:0090027","negative regulation of monocyte chemotaxis",-1.91373199255966,0.00014223524399644,
"G0:0036297","interstrand cross-link repair",-1.94198349103602,0.000154554427587156,0.0820684016
"G0:0031624","ubiquitin conjugating enzyme binding",-2.04276813475632,0.000165019831784404,0.083
"G0:0006303","double-strand break repair via nonhomologous end joining",-1.99692954559623,0.0002
"G0:0005657","replication fork",-2.02122506207052,0.000215042177450913,0.0000000000000000,"
```

```
"ID","Description","NES","pvalue","p.adjust","geneID","geneName","Count"
"KEGG_hsa03460","Fanconi anemia pathway",-1.95381165607489,0.000195517199959871,0.042818266791
"KEGG_hsa00450","Selenocompound metabolism",1.87533023860194,0.00148597618485593,0.16271439224
"KEGG_hsa03440","Homologous recombination",-1.77327611262785,0.00244617248053289,0.17857059107
"KEGG_hsa05134","Legionellosis",1.56257743506264,0.00890909482405167,0.379278158611733,"3329/5
"KEGG_hsa05030","Cocaine addiction",1.53845891460984,0.00918477245669554,0.379278158611733,"70
"KEGG_hsa04340","Hedgehog signaling pathway",-1.59107172801865,0.0103911824277187,0.3792781586
"KEGG_hsa00603","Glycosphingolipid biosynthesis",-1.59211334820907,0.0227929742569471,0.633647
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"KEGG_hsa05212","Pancreatic cancer",1.43290054501714,0.02604032363645,0.633647875153617,"1147/
"KEGG_hsa03060","Protein export",-1.5147581329971,0.0317635570566673,0.66197418743562,"11231/6
"KEGG_hsa03050","Proteasome",-1.41830306366179,0.0332498450310129,0.66197418743562,"5698/5694/
"KEGG_hsa05133","Pertussis",1.34273772074801,0.0576715515645589,0.986030939736781,"729238/3592
"KEGG_hsa04960","Aldosterone-regulated sodium reabsorption",-1.37388494309695,0.05964214711729
"KEGG_hsa04933","AGE-RAGE signaling pathway in diabetic complications",1.25193110216617,0.0719
"KEGG_hsa00600","Sphingolipid metabolism",1.3154921229844,0.0729613733905579,0.986030939736781
"KEGG_hsa04622","RIG-I-like receptor signaling pathway",1.30197191103829,0.0833333333333333,0.
"KEGG_hsa03450","Non-homologous end-joining",-1.44892228503064,0.0861723446893788,0.9860309397
"KEGG_hsa04744","Phototransduction",1.316518186593,0.104125736738703,0.986030939736781,"9187/6
"KEGG_hsa00920","Sulfur metabolism",1.42372425088798,0.105058365758755,0.986030939736781,"8991
"KEGG_hsa05220","Chronic myeloid leukemia",1.23508444609928,0.108647450110865,0.98603093973678
"KEGG_hsa00604","Glycosphingolipid biosynthesis",-1.3763081582424,0.110429447852761,0.98603093
"KEGG_hsa04614","Renin-angiotensin system",1.31791792676764,0.120792079207921,0.98603093973678
"KEGG_hsa01230","Biosynthesis of amino acids",1.25502165875818,0.122105263157895,0.98603093973
"KEGG_hsa05142","African trypanosomiasis",1.26455025022414,0.1220265070236508,0.986030939736781
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Thank you!



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